# Ultra Hard Final Exam – Machine Learning (MDS + BDMA + MIRI)

Duration: 1h30m

*All questions have equal weight. Justify all your answers. Be concise and precise.*

## Question 1

Let $X \in R^{n \times d}$ be a dataset with $n \ll d$. You apply logistic regression with L2 regularization.

(a) Explain in detail why the solution may still be unstable or poorly generalizing despite regularization.

(b) What would be the effect of using L1 regularization instead?

(c) Describe how you would use cross-validation to choose the regularization strength without overfitting to the validation set.

## Question 2

You are given a binary classification dataset where all features are highly non-Gaussian, and one of them dominates the rest in scale.

You want to compare the performance of: - Gaussian Naive Bayes - Logistic Regression - k-NN

For each model:

(a) Describe the assumptions violated by this dataset.

(b) Predict the likely behavior of the model and justify.

(c) Suggest a remedy for each model, if applicable.

## Question 3

You train a random forest with 200 trees. You observe: - 98% training accuracy - 75% cross-validated accuracy - 50% test accuracy on a new, external dataset

(a) What are three plausible causes for this pattern?

(b) How would you diagnose whether this is a result of overfitting, distribution shift, or data leakage?

(c) Suggest steps to address each of the three causes.

## Question 4

Given a neural network with sigmoid activation functions and no regularization:

(a) Explain why gradient descent training might fail, even with a good learning rate.

(b) Describe how feature scaling interacts with sigmoid units and training dynamics.

(c) Could switching to ReLU activations fix the issue entirely? Justify rigorously.

## Question 5

**Challenge:** Consider a dataset where: - Logistic regression gives near-perfect classification - A decision tree (even of large depth) performs poorly - Bagging the decision tree improves only slightly - A support vector machine with RBF kernel performs poorly

(a) Construct a theoretical explanation for this unusual model behavior.

(b) What is the underlying structure of the data that might lead to this?

(c) How would you empirically detect this kind of problem?

(d) Propose a new model or approach and justify your choice.